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Lecture 15: Optimal Transport for Graph Representation

1 Optimal Transport for Graph Representation

Graphs are everywhere. We have social networks (advertisements), brain connectivity (neuroscience), word relationships (NLP), recommender systems, 3D meshes (computer graphics), transportation networks, knowledge graphs (causality), gene regulatory network, neutrino detections. *These are all using graphs/networks.*

- Classical approach: Spectral and Fourier based analysis and processing (GNN)
- **Today:** Modeling graph as probability distributions using OT

2 Optimal Transport and divergences between graphs

2.1 Gromov-Wasserestein and Fused Gromov-Wasserstein

Definition 1 (GW for discrete distributions)**.** *[Mé11]*

$$GW_p^p(\mu_s, \mu_t) = \min_{T \in \Pi(\mu_s, \mu_t)} \sum_{i,j,k,l} |D_{i,k} - D'_{j,l}|^p T_{i,j} T_{k,l}$$

with $\mu_s = \sum_i a_i \delta_{x_i^s}$ *and* $\mu_t = \sum_j b_j \delta_{x_j^t}$, $D_{i,k} = \left\|x_i^s - x_k^s\right\|$ *and* $D' = \left\|x_j^t - x_k^t\right\|$

- Distance between metric measured spaces: across different spaces
- Search for an OT plan that preserves the pairwise relationships between samples.
- Entropy regularized GW proposed because GW is very sensitive [PCS16]
- Fused GW interpolates between Wass. and GW [VCF+18]

Definition 2 (Fused Gromov-Wasserstein (FGW) [VCF+18])**.**

$$FGW_p^p(\mu_s, \mu_t) = \min_{T \in \Pi(\mu_s, \mu_t)} \sum_{i,j,k,l} ((1-\alpha) C_{i.j}^q + \alpha \left|D_{i,k} - D'_{j,l}\right|^q)^p T_{i,j} T_{k,l}$$

with $\mu_s = \sum_i a_i \delta_{x_i^s}$ *and* $\mu_t = \sum_j b_j \delta_{x_j^t}$ *and* $D_{i,k} = \left\|x_i^s - x_k^s\right\|$, $D'_{j,l} = \left\|x_j^t - x_l^t\right\|$.

- Distance between metric measured spaced: across different spaces
- Search for an OT plan that preserve the pairwise relationships between samples
- Entropy regularized GW proposed in [PCS16]
- Fused GW between Wass. and GW [VCF+18]

2.2 Relaxing the marinals contrasints

Definition 3 (Unbalanced Gromov-Wasserstein [SVP20])**.**

$$\min_{T \in \Pi(\mu_s, \mu_t)} \sum_{i,j,k,l} |D_{i,k} - D'_{j,l}|^p T_{i,j} T_{k,l} + \lambda^u D_\varphi(\mathbf{T}^T \mathbf{1}_n, a) + \lambda^u D_\varphi(\mathbf{T}^T \mathbf{1}_n, b)$$

- The marginal constraints are relaxed by penalizing with divergence D_φ
- Partial GW proposed in [CAG20]
- Unbalanced FGW [TTZ+22] and Low rank [SKPC23]

Semi-relaxed (F)GW [VCFC$^+$22]

Definition 4.

$$\min_{T\geq 0, T1_m = a} \sum_{i,j,k,l} |D_{i,k} - D'_{j,l}|^p T_{i,j} T_{k,l}$$

- Second marginal constraint relaxed: optimal weights b wrt GW.
- Very fast solver (Frank-Wolfe) because constraints are separable

2.3 Graphs seen as distributions for GW

Take the tuple (D, F, h).

- The positions x_i are implicit and represented as the pairwaise matrix D
- Possible choices for D: Adjacency matrix, Laplacian, Shortest path
- The node features can be compared between graphs and stored in F
- h_i are the masses on the nodes of the graphs (uniform by default)

2.4 Applications

- Brain alignment between individual geometries using Fused unvlanaced Gromov-Wasserstein [TTZ$^+$22]
- Shape matching between surface with GW [SPKS16]
- GW alignment of word embedding spaces [AMJ18]

GW and extensions

- GW [Mé11] and FGW [VCF$^+$18] are versatile distances for graph and structed data seen as distributions
- Unbalanced [SVP20] and semi-relaxed [VCFC$^+$22]

GW tools

- OT plan gives interpretable alignment between graphs
- GW geometry allows barycenter and interpolation between graphs
- GW provides similarity between graphs (data fitting)

3 Learning graph representation with OT

3.1 GW barycenters and applications

Recall the Euclidean barycenter as $\min_x \sum_k \lambda_k \|x - x_k\|^2$. Now the FGW barycenter is

$$\min_{D\in\mathbb{R}^{n\times n}, \mu} \sum_i \lambda_i FGW(D_i, D, \mu_i \mu),$$

and it can

- Estimate FGW barycenter using Frechet means [PCS16]
- Barycenter optimization solved via block coordinate descent
- extension of K-means clustering FGW [VCF$^+$20]
- Use for data augmentation [MCW$^+$23]

3.2 Dictionary learning with OT

Represntation learning for graphs looks like

$$\min_{\{\bar{C}_k\}_k, \{\mathbf{w}_i\}_i} \frac{1}{N} \sum_i GW(\mathbf{C}_i, \hat{\mathbf{C}}(\mathbf{w}_i)).$$

- Learn a dictionary of graph templates to describe a continuous manifold
- The representation is learned by minimizing the FGW distance between the graph reconstruction form the embedding in the dictionary
- Online Graph Dictionary Learning: Linear model

$$\hat{\mathbf{C}}(\mathbf{w}) = \sum_k w_k \bar{C}_k$$

- GW Factorization: Nonlinear (GW barycenter) model [XLCZ20]

$$\hat{C}(w) = \operatorname*{argmin}_C \sum_k w_k GW(C, \bar{C}_k)$$

4 Supervised learning with OT on graphs

4.1 Graph classification with OT

Graph kernels and FGW:

- Graph kernels still SOTA on many datasets: WWL [TGLL$^+$19]
- FGW can vee used in a non-positive "kernel" [VCF$^+$20]
- Graph dictionary learning methods provide euclidean embeddings for kernels [VCFC$^+$22]

Graph Neural Networks [BBL$^+$16]

- Each layer of the GNN compute features on graph node using the values from the connected neighbors: message passing principle
- The final pooling step must remain invariant to permutations (min, max, mean)
- *Can we encode graphs as distributions in GNN?*

Template based FGW layer (TFGW)

- Principle: represent a graph through its distances to learned templates
- Novel pooling layer derived from OT distances
- New end-toend GNN models for graph-level tasks
- Compares well with state of the art GNN and graph kernel methods

Supervised graph prediction (a.k.a. graph regression)

- Objective: learn a function f predicting a graph g from an input x.
- Applications of SGP:
  - knowledge graph extraction [MDD22]
  - Natural language processing [DM16]
  - Molecule identification in chemistry [BSD+16]
- Surrogate based methods [BSD+16, ABMLdB23]
  - Represent graph as a vector in a high dimensional space
  - Learn a mapping from input to this space
  - Decode the vector to a graph
- Linear regression of Adjacency matrix [CFV22]

4.2 Structured graph prediction with OT barycenters and Any2Graph

Structured graph prediction with GW barycenters

$$f(\mathbf{x}) = \hat{\mathbf{C}}(\mathbf{w}(x)) = \arg\min_{\mathbf{C}} \sum_k w_k(\mathbf{x}) GW(\mathbf{C}, \bar{\mathbf{C}}_i)$$

- Predictions of the graph with a GW barycenter with weights conditioned by x
- Dictionary $\{\bar{C}_k\}_k$ and conditional weights w(x) learned simultaneously with

$$\min_{\{\bar{\mathbf{C}}_k\}_K, w(\cdot)} = \frac{1}{N} \sum_i GW(f(\mathbf{x}_i), \mathbf{C}_i)$$

- Both parametrix and non parametrix estimators [BMFB+22]
- Very powerful but slow at training and prediction due to barycenter computation.

Q: Combining this method with more normal OT methods? Like graphs exist in a space together, match both internal structure and position in space at the same time

A: That's what the fused GW is, there is a joint space and yea

Relationformer [SKW+22]

- Predict a graph o f max size M and activation scores for nodes to keep.
- Encoder-Decoder Transformer to predict node embedd
- Loss solves linear assignment problem (Hungarian) and uses assignment in quad loss between graphs of same size
- Fast prediction of graphs but focused on Image2Graph

Any2Graph Framework: Principle [KYF+24]

- End-to-end supervised graph prediction with a deep learning framework
- Learning optimization problem:

$$\min_{\theta} \frac{1}{n} \sum_{t=1}^{n} L(f_\theta(x_i), P(g_i)).$$

- $\{x_i, g_i\}$ are the input/output training data and P is a padding operator
- f_θ is a transformer
- f_θ also predicts is a padding vector $\hat{\mathbf{h}}$
- L is an optimal transport based loss for permutation invariant prediction

Q: Can you predict a special type of graph in an OT way? As in classical constraint, e.g. no cycles? A: That is an open question.

Any2Graph Neural network architecture

- The encoder extracts a set of features $x \longrightarrow (\mathbf{V}_1, \ldots, \mathbf{V}_k) \in \mathbb{R}^{k \times d}$
- The transformer translates them into M nodes embedding $(\mathbf{Z}_1, \ldots, \mathbf{Z}_M) \longrightarrow \mathbb{R}^{M \times d}$
- The decoder produces the graph following

$$\begin{aligned} \hat{h}_i &= \sigma(\mathrm{MLP}_m(\mathbf{z}_i)) \quad \forall i && \in \{1, \ldots, M\}, \\ \hat{F}_i &= \mathrm{MLP}_f(\mathbf{z}_i) \quad \forall i && \in \{1, \ldots, M\}, \\ \hat{A}_{i,j} &= \sigma(\mathrm{MLP}_s(\mathbf{z}_i + \mathbf{z}_j)) \quad \forall i, j && \in \{1, \ldots, M\}^2 \end{aligned}$$

- Similar to Relationformer [SKW+22] but with symmetric adjacency matrix

Partially-Masked Fused Gromov-Wasserstein (PM-FGW)

Definition 5 (PM-FGW)**.**

$$\textit{PM-FGW}(\hat{y}, y) = \min_{\mathbf{T} \in \Pi_M} L_{\mathbf{T}}(\hat{y}, y),$$

where

$$L_{\mathbf{T}}(\hat{y}, y) = \underbrace{\frac{\alpha_h}{M} \sum_{i,j} T_{i,j} \ell_h(\hat{h}_i, h_j)}_{\textit{padding loss}} + \underbrace{\frac{\alpha_f}{m} \sum_{i,j} T_{i,j} \ell_f(\hat{f}_i, f_j) h_j}_{\textit{feature loss}} + \underbrace{\frac{\alpha_A}{m^2} \sum_{i,j,k,l} T_{i,j} T_{k,l} \ell_A(\hat{A}_{i,k}, A_{j,l}) h_j h_l}_{\textit{structured loss}}$$

- ℓ_h, ℓ_f, ℓ_A are loss functions for node, feature and adjacency matrix discrepancies (KL when target discrete, Squared loss when continuous feature)
- $\alpha_h, \alpha_f, \alpha_A$ are hyperparameters on the simplex
- Loss is highly asymmetric due to the right masking by $\mathbf{h}$.
- Can be solved by Conditional Gradient with $\mathrm{O}(M^3 \log M)$ iteration

Any2Graph NN architecture It works extremely well!

5 Scaling graph OT solvers with neural networks

Challenges

- OT solvers iter. scale cubically with number of nodes
- Large graphs are too slow for many applications
- Approximate entropic solvers exist but still slow and dense OT plans are sub-optimal for graphs

Scaling OT on graphs with NN

$$\min_{\mathbf{T}} L_{OT}(\mathbf{T}, G, \hat{G}) \implies \min_{\theta} L_{OT}(\mathbf{T}_\theta, G, \hat{G})$$

- Learn to optimize with amortized optimization [Amo22]
- Predicting the OT plan for the large dataset of small graphs [noa]
- Prediction the Unbalanced OT plan between large graphs [MFT25]

$$\frac{1}{N}\sum_i \min_{\mathbf{T}} L_{OT}(\mathbf{T}, G^i, \hat{G}^i) \implies \min_{\theta} \frac{1}{N}\sum_i L_{OT}(\mathbf{T}_\theta(G^i, \hat{G}^i), G^i, \hat{G}^i)$$

5.1 GRAph Level autoEncoder (GRALE)

GRALE

- Train a Graph Level AutoEncoder: GRaph2Vec + Vec2Graph
- Build on ANy2Graph architecture for graph decoding
- Use node embeddings to predict OT plans and optimize PM-FGW loss
- Train simultaneously the Graphs encoder/decoder and the OT plan predictor
- Use Evoformer for graph encoding and decoding
- Train on large datasets of small graphs (coloring, Molecules)

GRALE Experiments

- GRALE outperforms state of art AE competitors on reconstruction and graph isomorphism accuracy
- GRALE scales to a large datasets of small graphs
- GRALE learns a latent space where interpolation/averaging is possible
- Embedding allows for semantic operations/editing on graphs
- Pre-trained GRALE encoder/decoder improves downstream graph tasks (regression, classification, graph prediction)

5.2 Unsupervised learning of OT plan prediction (ULOT)

ULOT for solving FUGW [MFT25]

$$\min_{\mathbf{T}\geq 0} \alpha \sum_{i,j,k,l} \left| D_{i,k} - D'_{j,l} \right|^2 T_{i,j} T_{k,l} + (1-\alpha) \sum_{i,j} C_{i,j} T_{i,j} + \rho(D(\mathbf{T}\mathbf{1}_m, a) + D(\mathbf{T}\mathbf{1}_n, b)).$$

- Learn to predict Unbalanced OT plan between large graphs
- Use graph NN and Attention layers to parameterize OT
- Optimize the FUGW loss over large dataset of graph pairs and parameters α, ρ:

$$\min_{\theta} xE_{\alpha,\rho,G,G'}[L^{\alpha,\rho}_{FUGW}(\mathbf{T}^{\alpha,\rho}_{\theta}(G,G'),G,G')]$$

- Provides after training a differentiable fast approximation of Unbalanced FGW for large graphs

ULOT numerical experiments

- Trained on datasets f simulated and fMRI brain graphs
- Efficient computation of continuous regularization path
- Differentiable OT layer w.r.t. both input graphs and FUGW parameters
- Correlation of 0.99 with exact FUGW loss on a test set
- Much faster than entropic OT approximation (100x) with similar performance
- Application on fMRI graph registration and prediction tasks

6 Conclusion

- Graphs modeled as distributions, GW can measure their similarity
- Extensions of GW for labels graphs and Frechet means can be computed
- Weights on the nodes are important but rarely available: relax the constraints [SVP20] or even remove one of them [VCFC$^+$22]
- Many applications of FGW for brain imagery [TTZ$^+$22] to Graph Neural Networks [VCFC$^+$22]
- OT is a powerful tool for (deep) graph structured prediction models [BMFB$^+$22, KYF$^+$24]
- Neural networks can help scale graph OT to large datasets or graphs [MFT25, noa]

7 Q & A

- Q: Much of what you talked about involved embeddings of graph into maps, can you speak a little on the identifiability question? There might be multiple graphs that correspond to this space.
- A: Designed the NN to deal with a lot of special input, that is why we used transformers. The decoder is fixed, you can rebuild the graphs by re-encoding.
- Q: When generating graph, do you specify number of nodes?
- A: Embedding captures size of graph and you can see continuous evaluation in distance space
- Q: What the topological properties of the graphs, are the relevant?

- A: Yes! Physical properties of the molecules are clearly encoded – some detail is inevitably lost, but joint structural property is encoded.
- Q: In the amortized setting when you predict the coupling – how long does that work outside the distribution?
- A: ULOT was applied on an extremely different structure and we only lost around 10%, of course not fully generalizable, but the more pairs of nodes in dataset the better it will perform.
- Q: Distance induces geometry on embedding space – so how do you validate that points in between are valid in geometry you choose rather than another?
- A: Refer to new paper on GRALE [noa]. You can see in the interpolation – it follows the constraints, if the chemistry and physics is in the constraints then it will be physically realistic. Sometimes it can learn the physical rules through the data and graph structure.
- Q: Once you learn the embedding space, have you tried using it for generative techniques?
- A: Yes you can do stable diffusion on graphs on this. It's a question of how good they will be. It's something we want to do but we don't have a very large team.

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